

Explore exoplanets yourself

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As part of the Science Year 2023 competition, pupils aged 15 and over had the opportunity to work with real observation data themselves. By analyzing light curves and spectra, they were able to determine some of the properties of the observed planets and thus gain a direct insight into the work of scientists. The competition was based on the transit and radial velocity methods used to search for exoplanets at the Wendelstein Observatory.

<i>How does astronomical research work?</i>	<i>2</i>
<i>Where does the data come from?</i>	<i>4</i>
<i>How do we analyze the data?</i>	<i>6</i>
<i>How can information about the star help and where can we find this information?</i>	<i>8</i>
<i>Task definition</i>	<i>9</i>
<i>But how do we get the mass of the planet?</i>	<i>9</i>
<i>Information on all exoplanets</i>	<i>12</i>
<i>Appendix</i>	<i>13</i>
<i>Direct installation of python miniconda3 on Windows</i>	<i>14</i>

How does astronomical research work?

Experimental research always consists of measurements and the subsequent interpretation of data.

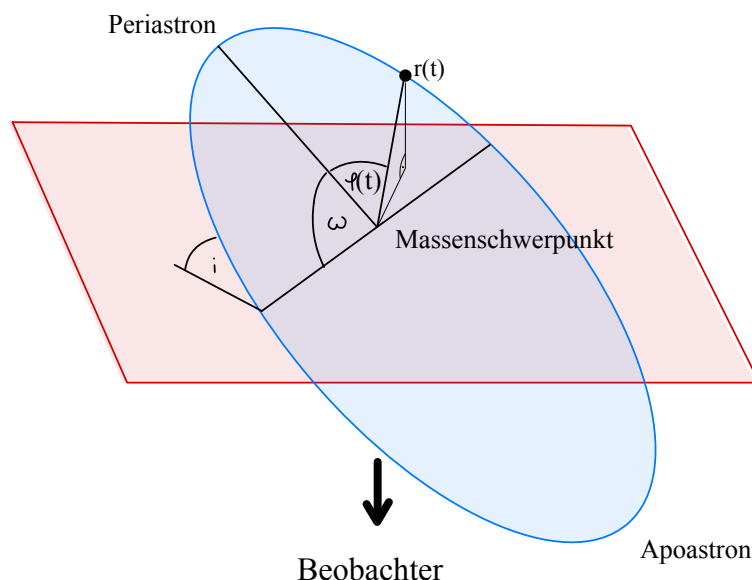
In our planetary research at the Wendelstein Observatory, it is changes in light on the one hand and changes in speed due to the Doppler effect on the other that are measured over a certain period of time. It is very practical that planets orbit their parent stars periodically and we can measure these changes again and again: The most direct (and also most precise) observation parameter is the **period** P . Since the astronomer Kepler, we have known the relationship between **the distance** between the planet and the star a and the **period** and, since Newton, also the relationship with the **mass of the star** M_s ¹:

$$a^3 \approx \frac{G}{4\pi^2} P^2 M_s$$

The planets revolve in an **elliptical orbit** (with the circle as a special case with eccentricity $e = 0$)

$$r(t) = \frac{a(1 - e^2)}{1 + e \cos \varphi(t)}$$

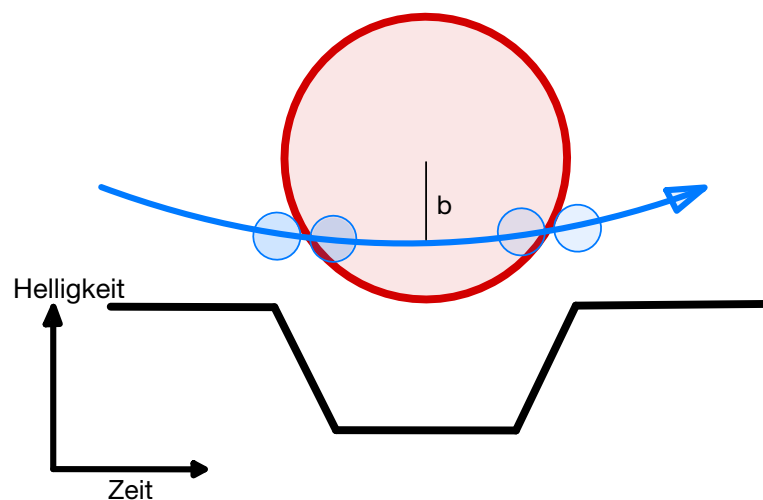
where $r(t)$ is the orbit as a function of time t , a is the major semi-axis of the ellipse, e is the eccentricity and $\varphi(t)$ is the angle of the orbit (see figure).



The inclination i and the argument of the periastron ω depend purely on the viewing direction of the observer.

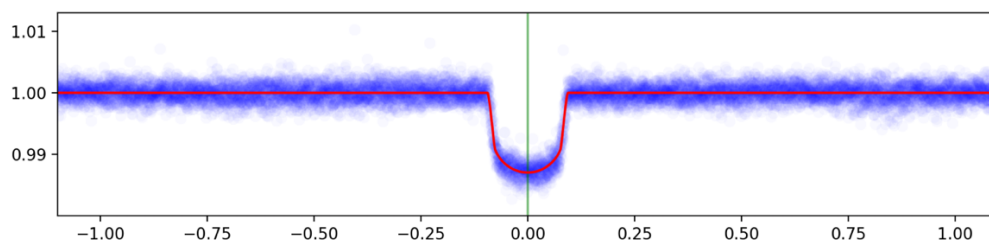
¹ Which researcher had previously discovered that the movement of small masses (planets) in the gravitational field (of the much more massive star) does not depend on the mass of the planet itself?

In order to detect exo-planets, i.e. planets outside our solar system, we measure the brief dimming when a planet passes in front of the star ("**transit**"). To do this, the orbit must be aligned exactly in our direction - the **inclination** i must have an angle of just under 90 degrees for this to happen. The planet then moves relatively evenly over the star's surface and leaves the star's disk again after a certain time. If you measure the brightness over a longer period of time as a function of the time of a star, the so-called "**light curve**", you can find such slight dimming.



Transit and light curve (black) with ingress and egress for an idealized uniformly bright star. The impact parameter b indicates how close the planet comes to the center of the star.

The **depth of the dimming** actually provides the size ratio between star and planet (R_p/R_s), the larger the planet the stronger the dimming. But even for the largest planets, this effect is very small: if we think of Jupiter, for example, which is 10 times smaller than the sun, the dimming would then only be 1% (since we are talking about surface area here), i.e. you have to measure very precisely!



Transit of TOI 0107 with all data from one period superimposed. The time axis the days relative to the time of transit, the brightness was measured relative to the basic brightness of the star. The dimming is slightly more than 1%.

A very important factor is the fact that the planet covers the star at all; this **probability** depends on the size of the star (the larger the more likely) and the distance between the planet and the star a (the further apart, the less the inclination angle may deviate from 90 deg!) However, it is very unlikely that a

transit will occur at all: in order to find an occultation by another planet similar to Jupiter in the same orbit around another sun at some point, you would have to monitor over 1000 stars for almost 12 years! If you observe for shorter periods, you have to be lucky that such a "transit" also takes place during the observation period, but this becomes increasingly unlikely with longer periods.

Where does the data come from?

We astrophysicists give luck a helping by using cameras that can record a large area of the sky to observe thousands of stars simultaneously over a longer period of time. NASA and MIT have a satellite called **TESS**² into orbit around the Earth, which permanently observes a large area of the sky (sector) for around 28 days at a time. Once a month, light curves of around 20,000 stars are published on a website³

However, the transits found in this data must then be verified by further observations to ensure that they were really caused by an exoplanet and not by something else. So as soon as a new transit is found in this data, it becomes interesting and the objects are then listed as **TOI**, the so-called "TESS Object of Interest". Immediately afterwards, many observatories around the world try to verify the transit, especially as the large pixels of TESS (which are necessary due to the large sky area) mean that it is not always clear whether a fainter neighboring star has generated the signal and whether it perhaps just a double star system after all.

In recent years, we have observed hundreds of such candidates (TOIs) at the Wendelstein Observatory in order to verify the suspected star as the correct one. Via the two internet addresses in the footnote you will find two of our light curves, which were recorded with the 43cm telescope at the Wendelstein observatory: one light curve of TOI 1420 in the infrared filter⁴ and the other of TOI 1877 in the red filter⁵. When you open the tables, the data appear in the first column as Julian date (days since a reference date), in the second column as normalized brightness, and in the third column as the associated error of the brightness measurement.

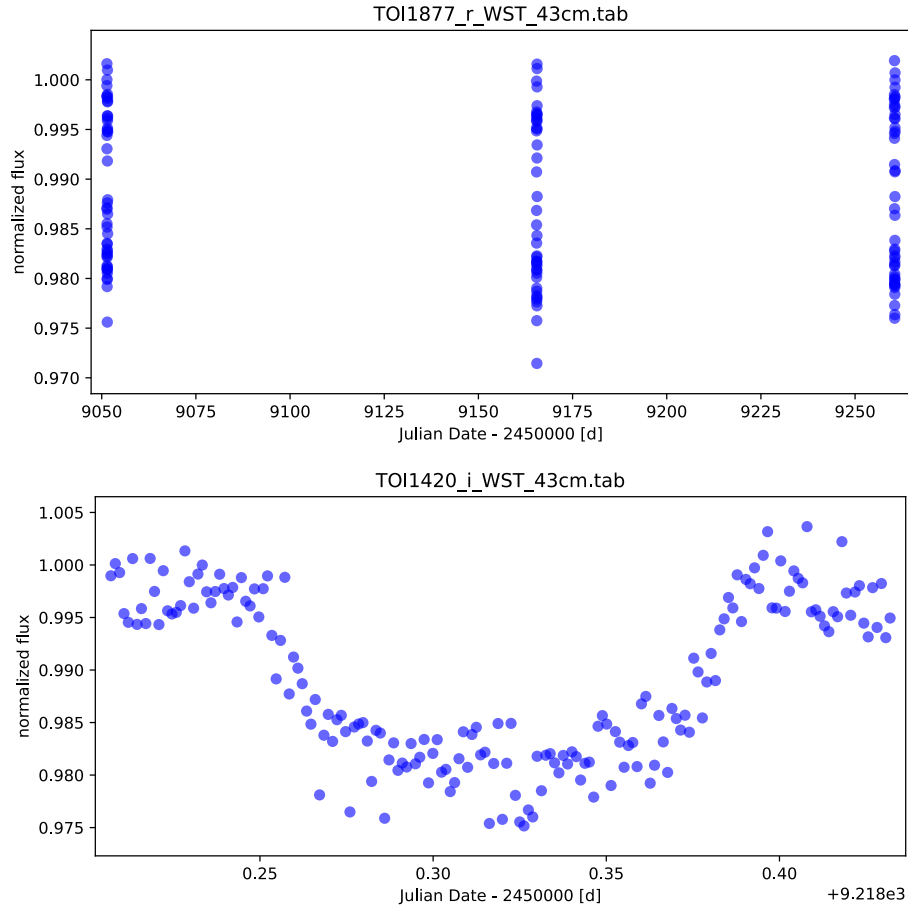
The transit was measured on three different nights for TOI 1877, but only on one night for TOI 1420. Although these light curves can already provide scientific results, the period cannot be determined clearly or at all. We should therefore also look at the original TESS data in order to supplement and compare the results.

² <https://tess.mit.edu>

³ https://archive.stsci.edu/tess/bulk_downloads.html

⁴ https://github.com/arriusm/planet_fast_fit/blob/main/examples/TOI1420/TOI1420_i_WST_43cm.tab

⁵ https://github.com/arriusm/planet_fast_fit/blob/main/examples/TOI1877/TOI1877_r_WST_43cm.tab



For all light curves recorded with TESS, the "Space Telescope Science Institute" (STScI) offers the possibility to download all data and light curves on its website⁶. As the download scripts are somewhat cryptic, we have provided a simple HTML file for download (download raw file here at [📄](#))

https://github.com/arriusm/planet_fast_fit/blob/main/TOI_lightcurves.html

which you can call up in a browser to retrieve any data from over 7000 TOIs via the corresponding links:

It is best to download the desired light curve by saving it under the simpler name displayed there. This gives you a better overview of many light curves.

⁶ https://archive.stsci.edu/tess/bulk_downloads/bulk_downloads_ffl-tp-lc-dv.html

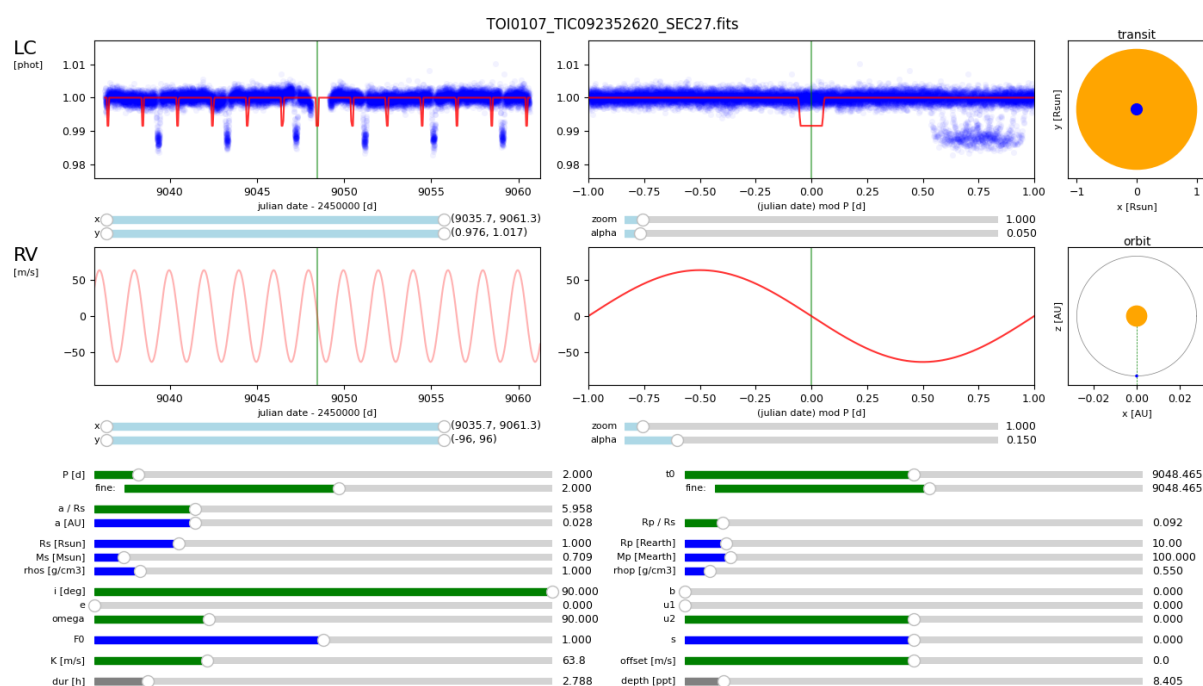
How do we analyze the data?

60 years ago, the work of an astrophysicist analyzing data could still be imagined at a desk with paper and pencil. Today, analysis with the help of computer programs is . Many scientists use *the Python programming language*, which is also commonly used in schools and simplifies the necessary calculations thanks to its many libraries. For example, we use the following 5 libraries to analyze planetary light curves: *numpy*⁷, *matplotlib*⁸, *argparse*⁹, *astropy*¹⁰ and *batman*¹¹. The library "*batman*" enables very precise predictions of how a transit must look depending on the elliptical planetary orbit. In the appendix you will find instructions on how to install *Python* and the libraries on your computer. So that you can analyze your light curves, we have prepared a *Python program* for you with which you can interactively measure the light curves:

https://github.com/arriusm/planet_fast_fit/blob/main/planet_fast_fit.py

You can use *the Python program* provided to display these light curves and try to adjust the parameters yourself using your own model, e.g. TOI 107, which can found in the data from Sector 27:

`python planet_fast_fit.py TOI0107_TIC092352620_SEC27.fits`



⁷ <https://numpy.org/>

⁸ <https://matplotlib.org/>

⁹ <https://docs.python.org/3/library/argparse.html>

¹⁰ <https://www.astropy.org/>

¹¹ <https://lkreidberg.github.io/batman/docs/html/index.html>

do this, you can move the 'sliders' to adjust the predicted transit light curve to the observed data.

The quickest way is to determine a **transit time** t_0 very precisely (several are possible).

You will also see that the **period** is also very easy to determine. For example, the period for TOI 1877 can be better determined by the much larger time span of the three observations at Wendelstein, which are more than a hundred days apart, if you take the period found with TESS data as the initial value and then finely adjust the period.

The light curve is a very good way of measuring how much time the planet needs to fly over the star. If the period is divided by these time intervals, the circumference and thus also the **distance between the star and the planet** can only be calculated in units of the stellar radius a/R_s

However, this is only possible if we know at which secant ('**impact parameter**' b) the planet has passed over the star, as the duration of the transit also depends on this. This is where the researchers help themselves with their knowledge of what the **center-edge dimming** of the star looks like: This influences in particular the shape of the flanks in the light curve in the entrance and exit, i.e. the point in time when the planet passes over the edge of the star

The **shape of the entrance and exit** also on other parameters, such as the brightness profile of the star, the so-called limb darkening. It turns out that this curve can be adjusted very well with a second-degree polynomial, i.e. it can be varied with two parameters (u_1, u_2), which parameterize the radial brightness profile of the star. This allows the impact parameter to be estimated and the ratio of the semi-major axis of the planetary orbit to the stellar radius a/R_s to be determined.

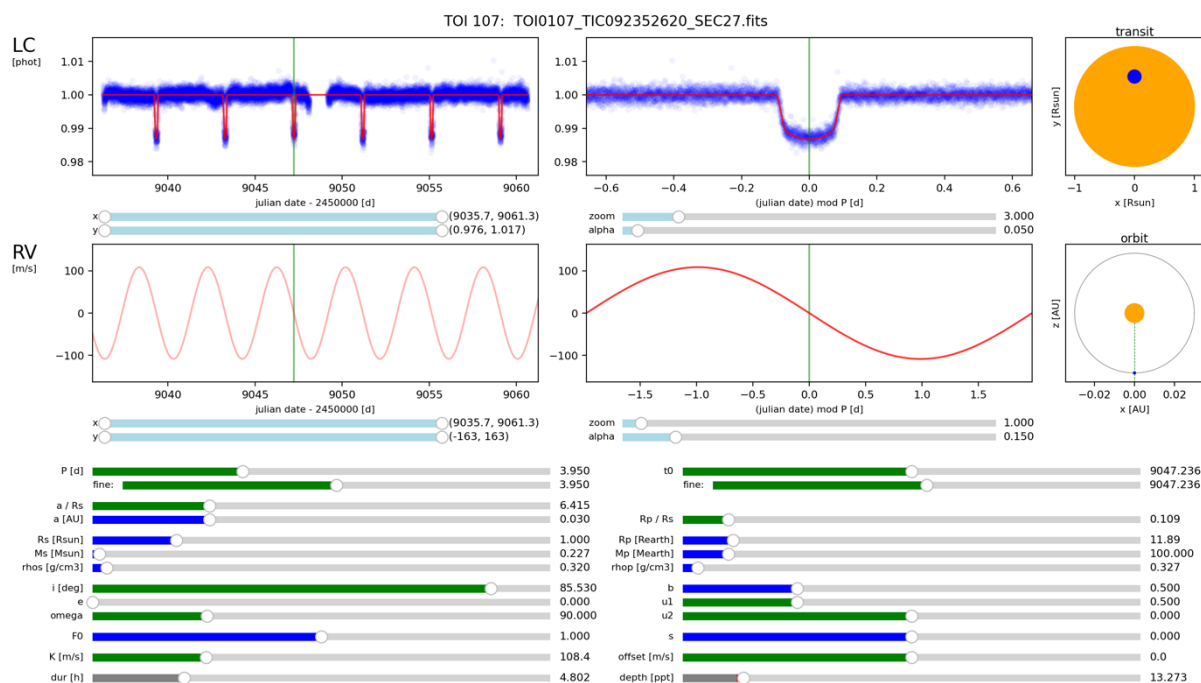
Of course, it also plays a role how **elliptical** the planet's orbit around the star is (eccentricity e) and how this ellipse is aligned as seen from us (parameter ω), e.g. whether we observe the transit rather in periastron (= point on the planet's orbit, which is closest to the star) or in apoastron (= point on the planet's orbit which is furthest away from the star) or somewhere in between: according to Kepler's 2nd law, this changes the speed at which the planet appears to pass over the star and this changes the duration of the transit. For our estimates here, we always take a circular orbit as a first approach, but we also know that eccentricity can change things.

With the **parameters** t_0 , s and F_0 the model can be moved in x and y direction and also scaled slightly.

One can consider that the knowledge of a/R_s together with the period P determines the **mean density of the star** (neglecting the planetary mass) if one uses Kepler's 3rd law (as above). The mean stellar density is therefore the link between the period and the relative orbital axis a/R_s

How can information about the star help and where can we find this information?

Therefore, knowing the density of the parent star can also help with the impact parameter. For TOI 0107 (TIC 92352620), a density of 0.32 g/cm^2 is given in the TIC catalog¹², which limits the range for the impact parameter b relatively well in our Python *program*. Bear in mind, however, that this value is subject to error!



Here we show you where to find the data and how to get the available information of the star for each star with a TIC number, because the star parameters (radius, mass) are crucial to then find out the parameters of the orbiting exo-planet:

<https://exofop.ipac.caltech.edu/tess>

On this page we are particularly interested in the stellar parameters with the radius (R_{Sun}), the mass (M_{Sun}) and the density (g/cm^3) with their errors ($\pm$). With regard to the errors, it should be noted that these parameters have not been determined with arbitrary accuracy: Radius 3-10 %, Mass 10-15 %, Density 15-25 %.

¹² <https://exofop.ipac.caltech.edu/tess/target.php?id=92352620>

With this information, the **radius of the planet in relation to the stellar radius** (R_p/R_s) can now also be found relatively precisely .¹³

Task definition

Using the Wendelstein data and the TESS data from TOI 1877 and TOI 1420, you should approximate the optimum parameters as closely as possible. Some parameters are dependent on others (which ones?), these are called degenerate. Also try to find out the possible inaccuracies and discuss them. How do the measurements differ between TESS data and Wendelstein?

Look for at least two other planet candidates of your choice from the TESS list and find out the parameters!

But how do we get the mass of the planet?

Here we use the mutual influence of star and planet. Not only does the planet orbit around the star, but the star also orbits around the common center of gravity. However, due to the much larger mass of the star, it only wobbles back and forth, always in the opposite direction to the planet.

We can actually measure this movement using the **Doppler effect**, spectral lines that appear at very specific points in a star's spectrum shift when the star changes its movement: if it moves toward us, they are blue-shifted, if the star moves away from us, they move in the reddish direction, i.e. the wavelength increases. So if you observe a star for a long time and measure these shifts in its spectrum, you can find out how this star is moving relative to us. From this we can then prove the existence of one or more planets. This method is called the **radial velocity method** (hence the RV method). Since even large, closely orbiting planets only cause a wobble of a few hundred meters per second at best, the wavelength only shifts by a tiny fraction, or more precisely, by the ratio between this speed and the speed of light, i.e. by less than one 3 millionth.

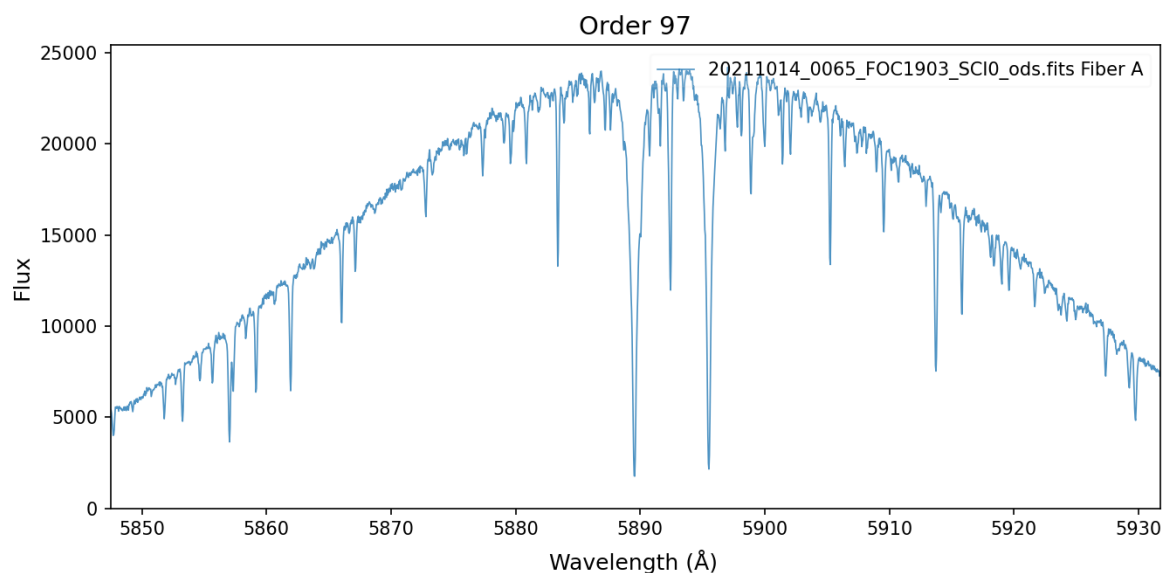
How can such a shift be measured when the resolution of the best spectrographs is only in the range of $1/100,000$ ¹⁴ ? Two things help: on the one hand, the center of a line is much better defined than the width of the line would suggest; on the other hand, many lines can be used together to measure the shift, which greatly improves the accuracy.

¹³ The parameters in the program can also be specified directly via options, e.g.
`python planet_fast_fit.py -rhostar 0.32 -P 3.9502 -t0 9047.236 -b 0.5 -rp 0.109 -u1 0.5 -zoom 3 -alpha 0.05 -save TOI0107_SEC27.pdf TOI0107_TIC092352620_SEC27.fits`

¹⁴ At the Wendelstein Observatory, the accuracy is $1/R=1/70000$.

We can try out the measurement of the parameters ourselves using the example of the star 51 Pegasi. It is a sun-like star with 1.09 solar masses, located 50.6 light years from Earth in the constellation Pegasus. It was the first main sequence star (star with hydrogen burning in its center) where an exoplanet was found in 1995 by Michel Mayor and Didier Queloz. In 2019, both were awarded the Nobel Prize in Physics for their discovery. The planet itself is designated as the second object (alongside the star) with 51 Peg b or "Dimidium".

In the following image we see a small part of the spectrum of the star 51 Peg, which was recorded with the Wendelstein 2.1 m telescope. The sodium D line (designated "D" by Joseph von Fraunhofer in 1814), which forms a doublet (2 lines) at the wavelengths 589.0 nm and 589.6 nm, is striking. With a change in motion of the star of 120 m/s, the lines move back and forth in their wavelength by approx. two ten-thousandths of a nanometer (0.0002 nm), which would not even be visible in this graphic!



Since 51Peg is one of our reference objects at the Wendelstein Observatory, we have already recorded it during many nights. Under the following link you will find a table with measurements from 14 nights between 9.10.2021 and 14.11.2021.

https://github.com/arriusm/planet_fast_fit/blob/main/examples/51Peg/51Peg_RV_MaHPS.dat

For the RV method, we have slightly extended our *Python* program: It also allows you to display and analyze the RV data¹⁵. This time it requires the *Python* library *radvel*¹⁶ to display radial velocity models.

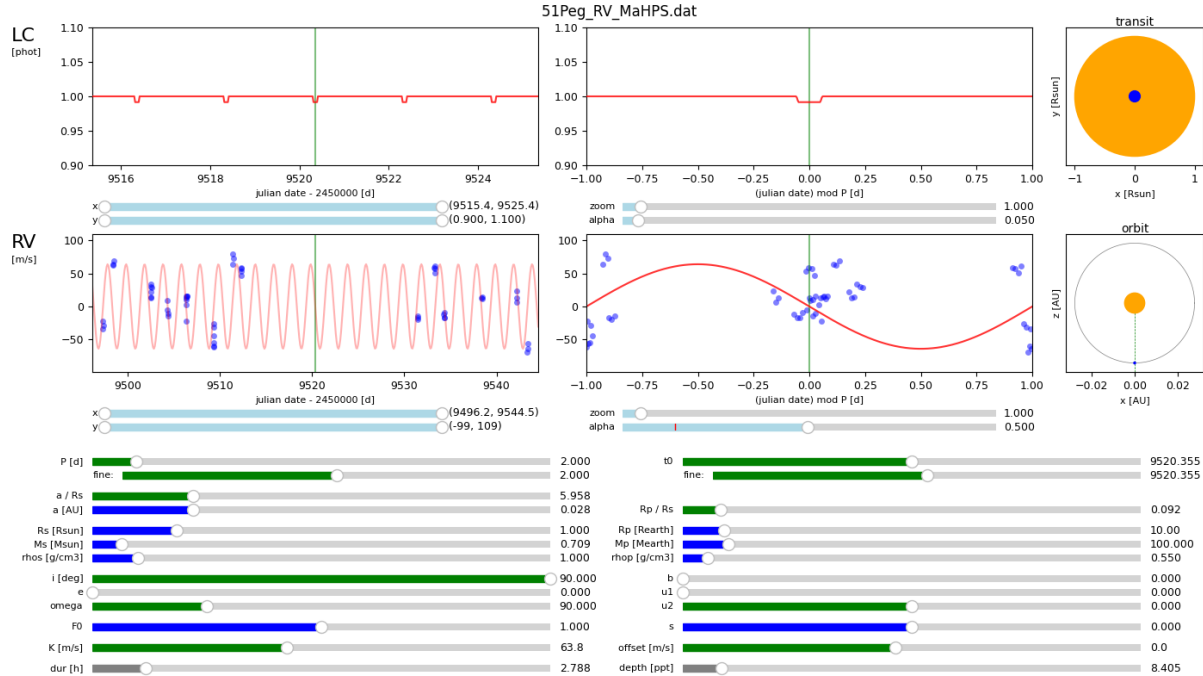
This time you can call up the program with the following command:

¹⁵ https://github.com/arriusm/planet_fast_fit/blob/main/planet_fast_fit.py

¹⁶ <https://radvel.readthedocs.io/en/latest/>

python planet_fast_fit.py -rv 51Peg_RV_MaHPS.dat

In the lower left-hand graph we show the entire RV curve, in the graph to the right of it we show the period-cut curve, .e. time points of the same phase appear one above the other.



The decisive parameter is K , the amplitude of the radial velocity. It can be calculated that it depends on the mass of the planet:

$$K \approx \sin(i) \frac{M_{\text{Planet}}}{M_{\text{Stern}}} \left(\frac{2\pi G M_{\text{Stern}}}{P} \right)^{1/3} \frac{1}{\sqrt{1 - \epsilon^2}}$$

However, since we do not know the inclination i of stars that do not show direct transit, the mass of the planet can only be determined with a pre-factor "sin(i)".

Task definition:

Try to the value for $M_{\text{planet}} \sin(i)$ for 51 Peg b. Which planet in our solar system is 51 Peg b most similar to, assuming an inclination of 80° ? At what distance does the planet orbit the star 51 Peg?

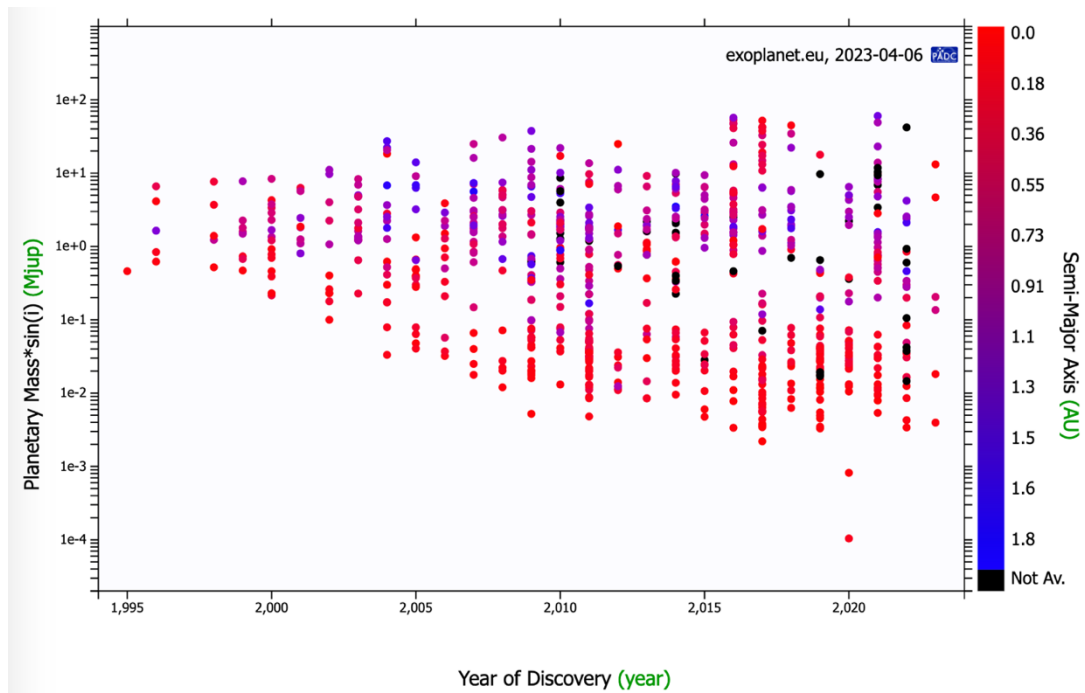
Information on all exoplanets

On the website

<http://exoplanet.eu>

you can get an overview of when which planets discovered using different methods

Here is an overview of the planets discovered



Where can we find 51 Peg

Today, it is standard procedure to subsequently analyze transit discoveries using the RV method and thus measure the most important planetary parameters!

Appendix

There are various freely available distributions for the *Python* programming language. You can install individual libraries for all of them. It's best to get help from a teacher...

Conda

<https://conda.io/projects/conda/en/stable/user-guide/install/index.html>

Anaconda: <https://www.anaconda.com/products/distribution>

Miniconda: <https://docs.conda.io/en/latest/miniconda.html>

```
conda install numpy
conda install matplotlib
conda install astropy
pip3 install argparse
pip3 install h5py
pip3 install celerite
pip3 install radvel
pip3 install batman-package
```

python.org

<https://www.python.org/downloads>

```
pip3 install numpy
pip3 install matplotlib
pip3 install astropy
pip3 install argparse
pip3 install h5py
pip3 install celerite
pip3 install radvel
pip3 install batman-package
```

h5py and *celerite* are required for *radvel*. All libraries are downloaded to a specific directory and installed here.¹⁷

¹⁷ e.g. on a macOSX operating system in
~/miniconda3/lib/python3.10/site-packages or
/Library/Frameworks/Python.framework/Versions/3.11/lib/python3.11/site-packages

Direct installation of python miniconda3 on Windows

(without WSL and Linux)

tested with Windows 11

i) miniconda3 installation:

Download the latest miniconda python 3.11 version in a "Windows PowerShell" in the home directory

```
curl https://repo.anaconda.com/miniconda/Miniconda3-py311\_24.11.1-0-Windows-x86\_64.exe -o miniconda.exe
```

and install

```
Start-Process -FilePath ".\miniconda.exe" -ArgumentList "/S" -Wait
```

Information on this can be found at:

<https://docs.anaconda.com/miniconda/install/#quick-command-line-install>

You will then see an "Anaconda Powershell Prompt (miniconda3)" in the start menu, which must be opened.

The simple version of python should run, simply enter "python" and exit with "exit()".

ii) Compiler:

Then comes an important thing: to install certain libraries, you need a compiler and some tools:

You can find them under

<https://visualstudio.microsoft.com/de/visual-cpp-build-tools>

with the Visual Studio Installer, installed the following

e.g. the "python native" packages are included.

1)

Visual Studio Build Tools 2022

Vers. 17.12.3

2)

Visual Studio Community 2022 (2) 417 packages!!!

Vers. 17.12.3

iii) python libraries

After installing the compiler, it is better to reopen the "Anaconda Powershell Prompt (miniconda3)" and then install the necessary libraries with pip:

```
pip install h5py
pip install celerite
pip install radvel
pip install windows-curses
pip install batman-package
```

Then in

```
import radvel
```

```
import batman
```